

Retail Sales Data Analysis

1. Introduction

The objective of this project is to analyze retail sales data and identify customer purchasing behavior, product performance, and sales trends. Exploratory Data Analysis (EDA) techniques were used to extract meaningful business insights from the dataset.

2. Dataset Description

The dataset contains retail transaction information including customer demographics, product categories, quantity purchased, pricing, and total sales amount.

Dataset Information

- Total Records: 1000
- Total Features: 9
- Missing Values: 0
- Duplicate Records: 0

Features

- Transaction ID
- Date
- Customer ID
- Gender
- Age
- Product Category
- Quantity
- Price per Unit
- Total Amount

3. Data Preprocessing

The following preprocessing steps were performed:

- Loaded dataset using Pandas
- Checked dataset structure using info()
- Verified missing values
- Checked duplicate records
- Converted Date column into datetime format
- Created Month feature for trend analysis

4. Exploratory Data Analysis

Gender Distribution

Analysis showed a balanced distribution between male and female customers.

Product Category Analysis

Product categories such as Clothing and Electronics generated significant sales activity.

Age Distribution

Customers belong to different age groups, indicating a diverse customer base.

Monthly Sales Trend

Monthly sales varied throughout the year, showing seasonal fluctuations in customer purchases.

Sales by Product Category

Revenue comparison highlighted the best-performing product categories.

Top Customers

A small group of customers contributed significantly to overall sales revenue.

Correlation Analysis

The correlation heatmap indicated a positive relationship between quantity purchased and total sales amount.

5. Business Insights

1. Clothing and Electronics contribute significantly to overall revenue.
2. Monthly sales patterns indicate seasonal demand fluctuations.
3. Customer purchasing behavior varies across age groups.
4. Top customers generate a substantial portion of total sales.
5. Higher quantities purchased generally result in higher revenue.

6. Recommendations

1. Increase inventory for high-performing product categories.
2. Introduce customer loyalty programs.
3. Run promotional campaigns during low-sales months.
4. Use demographic insights for targeted marketing.
5. Focus marketing efforts on top-selling categories.

7. Conclusion

The analysis successfully identified important sales trends, customer behavior patterns, and revenue drivers. These insights can help businesses improve inventory planning, marketing strategies, and customer retention efforts.